History of Artificial Intelligence & Deep Learning

Vatsav Reddy

Under the Guidance of
Prof. Praveen Parchuri

International Institute of Information Technology, Hyderabad

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Course Overview

Topics Covered

- Deep Generative Models
- Restricted Boltzmann Machines
- Variational Autoencoders (VAEs)
- Autoregressive Models
- Generative Adversarial Networks (GANs)

Main Architectures

CNNs, RNNs, LSTMs, Attention Mechanisms

Reference

J. Schmidhuber, *Deep Learning: An Overview*

Reticular Theory vs. Neuron Doctrine

Reticular Theory (1871–1873)

- Proposed by Joseph von Gerlach
- Nervous system viewed as a single continuous network
- Camillo Golgi developed staining techniques
- Golgi supported Reticular Theory

Neuron Doctrine (1888–1891)

- Proposed by Santiago Ramón y Cajal
- Nervous system made of discrete neurons
- Neurons communicate through synapses
- Term “neuron” coined by Waldeyer-Hartz

1906 Nobel Prize

Golgi and Cajal jointly received the Nobel Prize in Physiology/Medicine despite supporting opposing theories.

The Final Word

Electron Microscopy (1950s)

Electron microscopy conclusively confirmed the Neuron Doctrine by showing that neurons are distinct individual cells connected through synapses.

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McCulloch-Pitts Neuron (1943)

- Proposed by Warren McCulloch and Walter Pitts
- Simplified mathematical model of a biological neuron
- Inputs: $x_1, x_2, \ldots, x_n$
- Output: binary value $y \in \{0, 1\}$

Threshold Logic

$$y = f\left(\sum_{i=1}^{n} w_i x_i\right) = \begin{cases} 1 & \text{if } \sum w_i x_i \geq \theta \\ 0 & \text{otherwise} \end{cases}$$

The Perceptron (1957–1958)

- "Artificial Intelligence" term coined in 1956
- Frank Rosenblatt proposed the Perceptron
- Inputs connected using learnable weights
- Could learn simple decision boundaries
- Received enormous media attention

Historical Quote

"The perceptron may eventually be able to learn, make decisions, and translate languages."

First Multilayer Perceptrons

Ivakhnenko et al. (1965–1968) proposed early multilayer neural networks.

Perceptron Limitations and AI Winter

Minsky & Papert (1968)

- Perceptrons cannot solve XOR
- XOR is not linearly separable
- Funding for connectionist AI decreased
- Led to the first AI Winter

Backpropagation (1986)

- Rediscovered several times during 1960s–1970s
- Paul Werbos used it for neural networks in 1982
- Popularized by Rumelhart, Hinton, and Williams (1986)

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From AI Winter to Deep Revival

- 1847: Cauchy introduced Gradient Descent
- 1989: Universal Approximation Theorem
- 1989–2006: Second AI Winter

Universal Approximation Theorem

A feedforward neural network with a single hidden layer can approximate any continuous function to arbitrary precision.

Deep Revival (2006)

Hinton and Salakhutdinov introduced effective pretraining methods for deep autoencoders.

Unsupervised Pretraining and GPU Era

- 1991–1993: Schmidhuber explored unsupervised pretraining
- 2007–2009: Greedy layer-wise pretraining became popular
- 2009: Graves et al. achieved Arabic handwriting recognition
- 2010: Dahl et al. reduced speech recognition error:

$$23.2\% \rightarrow 16.0\%$$

- GPUs dramatically accelerated backpropagation
- 2011: Traffic sign recognition achieved 0.56% error rate

ImageNet Challenge

Network	Top-5 Error	Layers
AlexNet	16.4%	8
ZFNet	11.2%	8
VGGNet	7.3%	19
GoogLeNet	6.7%	22
ResNet (2015)	3.6%	152

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Hubel and Wiesel Experiment (1959)

Key Discovery

Each neuron responds only to stimuli in a specific receptive field of visual space.

- "Neurons that fire together wire together"
- Inspired hierarchical visual processing models

Neocognitron and CNNs

Neocognitron (1980)

Kunihiko Fukushima proposed the Neocognitron for handwritten character recognition.

- Late 1980s–1990s: CNNs popularized by Yann LeCun
- LeNet-5 (1998) achieved success on MNIST
- CNNs later enabled large-scale image recognition

Max Pooling

$$J(x,y) = \max_{m,n} I(x+m, y+n)$$

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Better Optimization Techniques

Optimization Advances (2012–2016)

- Faster convergence
- Better generalization
- More stable training
- Nesterov Momentum improved SGD

Year	Optimizer
2011	AdaGrad
2012	RMSProp
2014	Adam
2015	Eve
2018	Beyond Adam

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Why Sequences Matter

Sequences Are Everywhere

- Time series
- Speech
- Music
- Text
- Video

Hopfield Networks (1982)

Associative memory systems capable of recalling patterns from noisy inputs.

Jordan and Elman Networks

Jordan Network (1986)

Output state is fed back into future time steps.

Elman Network (1990)

Hidden state is propagated across time steps.

RNN Problems and LSTMs

Exploding and Vanishing Gradients

Hochreiter and Bengio demonstrated the difficulty of training RNNs over long sequences.

Long Short-Term Memory (LSTM) (1997)

LSTMs solved long-term dependency problems using gated memory cells.

Sequence-to-Sequence Learning

- Sutskever et al. (2014)
- Bahdanau et al. (2015) introduced Attention

Attention and Reinforcement Learning

Schmidhuber & Huber (1991)

Used reinforcement learning to determine where a model should focus attention.

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AI Beats Humans at Games

- 2015: Deep Q-Networks (DQN) mastered Atari games
- 2016: AlphaGo defeated world champion Lee Sedol
- 2016: DeepStack defeated professional poker players
- 2017: OpenAI Five played Defense of the Ancients 2 (Dota 2)

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Language Modeling and Translation

Language Modeling

- Mikolov et al. 2010
- Kiros et al. 2015
- Kim et al. 2015

Machine Translation

- Kalchbrenner et al. 2013
- Cho et al. 2014
- Bahdanau et al. 2015
- Luong et al. 2015
- Sutskever et al. 2014

Speech Recognition

- Hinton et al. 2012
- Graves et al. 2013
- Chorowski et al. 2015
- Sak et al. 2015

Conversation and Question Answering

Conversation Modeling

- Shang et al. 2015
- Vinyals et al. 2015
- Lowe et al. 2015
- Serban et al. 2016
- Bordes et al. 2017

Question Answering

- Hermann et al. 2015
- Chen et al. 2016
- Xiong et al. 2016
- Dhingra et al. 2017
- Wang et al. 2017

Object Detection and Recognition

- Semantic Segmentation (Long et al. 2015)
- Faster R-CNN (Ren et al. 2015)
- Inside-Outside Net (Bell et al. 2015)
- YOLO9000 (Redmon et al. 2016)
- R-FCN (Dai et al. 2016)
- Mask R-CNN (2017)
- Video Object Segmentation (Caelles et al. 2017)

Visual Question Answering

- Agrawal et al. 2015
- Yu et al. 2017
- Johnson et al. 2017
- Ben-Younes et al. 2017
- Malinowski et al. 2017
- Kazemi et al. 2017

Video Question Answering

- Tapaswi et al. 2016
- Zeng et al. 2016
- Zhao et al. 2017
- Jang et al. 2017
- Xu Hongyang et al. 2017
- Maharaj et al. 2017

Image and Video Captioning

Image Captioning

- Mao et al. 2014
- Kiros et al. 2015
- Donahue et al. 2015
- Vinyals et al. 2015
- Karpathy et al. 2015

Video Captioning

- Donahue et al. 2014
- Venugopalan et al. 2014
- Pan et al. 2015
- Yao et al. 2015
- Rohrbach et al. 2015

Generative Models

Generating Realistic Photos

- Variational Autoencoders — Kingma et al. 2013
- GANs — Goodfellow et al. 2014
- Plug and Play Generative Networks — Nguyen et al. 2016
- Progressive Growing of GANs — Karras et al. 2017

Generating Audio

WaveNet — van den Oord et al. 2016

Transformers (2017)

Attention Is All You Need

Vaswani et al. introduced the Transformer architecture based entirely on self-attention mechanisms.

- Removed recurrence
- Highly parallelizable
- Became foundation of modern LLMs

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Why Does Deep Learning Work So Well?

Open Problems

- High capacity and overfitting
- Numerical instability
- Exploding and vanishing gradients
- Sharp minima
- Adversarial examples

Current Direction

- Explainability
- Robustness
- Theoretical understanding
- Reliable AI systems

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Timeline Summary

- 1871 – Reticular Theory
- 1888 – Neuron Doctrine
- 1943 – McCulloch-Pitts Neuron
- 1957 – Perceptron
- 1968 – AI Winter
- 1980 – Neocognitron
- 1986 – Backpropagation
- 1989 – Universal Approximation Theorem
- 1997 – LSTM
- 2006 – Deep Revival
- 2012 – AlexNet and GPU Revolution
- 2014 – Seq2Seq and GANs
- 2015 – DQN and ResNet
- 2016 – AlphaGo
- 2017 – Transformers

Thank You

Questions?